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Bonding device

Abstract

A bonding device is used to bond a plate assembly including a first plate and a second plate and includes an anvil, a probe, a shoulder component, a driving mechanism, a power supply, and a control part. The anvil is configured to support a first surface of the plate assembly provided by the first plate. The probe is disposed at a position corresponding to the anvil to be opposite to a second surface of the plate assembly provided by the second plate. The shoulder component is disposed around the probe and has a via for the probe to pass through and an abutting surface for pressing the second surface. The anvil has a hemispherical surface facing the first surface and a receiving surface disposed at a position corresponding to the abutting surface. The anvil contacts the first surface with the hemispherical surface and the receiving surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims the priority benefit of China application serial no. 202311164540.9, filed on Sep. 11, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.
BACKGROUND
Technical Field
(2) The disclosure relates to a bonding device.
Description of Related Art
(3) In the prior art, resistance welding and friction stir welding may be listed as methods for

bonding multiple metal plates, which can suppress gas generation more than conventional bonding methods to reduce the negative environmental impact on air quality. In addition, the friction stir welding consumes less electricity than the conventional bonding methods, contributing to improving energy efficiency during manufacturing. When three or more metal plates made of different materials are bonded through the resistance welding, it is difficult to maintain stable bonding strength for a long time. The friction stir welding is limited to the range of materials that can be effectively stirred and is not suitable for bonding three or more metal plates. Therefore, a bonding technology for bonding three or more metal plates by simultaneously using the resistance welding and the friction stir welding is developed (for example, Japanese Patent Publication No. 2023-013804).

(4) In such a bonding device, when a plate assembly composed of multiple metal plates is placed between an anvil and a probe of the bonding device to be bonded via energization, two ends of the plate assembly may be easily bent and deformed by the abutment of shoulder components disposed around the probe, thereby increasing a conductive contact area (that is, expanding a current flow path) between the bent and deformed plate assembly and the anvil. At this time, it may be difficult to generate a nugget for bonding, which may cause difficulty in improving the bonding effect.

(5) The disclosure provides a bonding device that can prevent the plate assembly from being bent and deformed to suppress an increase in the conductive contact area, thereby improving the bonding effect. Furthermore, it is expected to help reduce the environmental impact on air and improve energy efficiency during manufacturing.

SUMMARY

(6) The disclosure provides a bonding device, which is used to bond a plate assembly including a first plate and a second plate configured to abut against the first plate and includes: an anvil, configured to support a first surface of the plate assembly provided by the first plate; a probe, disposed at a position corresponding to the anvil to be opposite to a second surface of the plate assembly provided by the second plate; a shoulder component, disposed around the probe and having a via for the probe to pass through and an abutting surface for pressing the second surface; a driving mechanism, configured to rotate the probe about a central axis intersecting the second surface and to move the probe forward or backward relative to the second plate along the central axis; a power supply, electrically connected to the anvil and the shoulder component, so that current flows through the plate assembly between the anvil and the shoulder component; and a control part, controlling operations of the driving mechanism and the power supply. The anvil has a hemispherical surface facing the first surface and being at least partially conductive and a receiving surface disposed at a position corresponding to the abutting surface of the shoulder component and being at least partially insulated, and the anvil contacts the first surface with the hemispherical surface and the receiving surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a cross-sectional schematic diagram of a bonding device for bonding a plate assembly including a first plate and a second plate according to an embodiment of the disclosure.

(2) FIG. **2** is a partially enlarged schematic diagram of the bonding device shown in FIG. **1** during a bonding process.

(3) FIG. **3** is a cross-sectional schematic diagram of an anvil used in the bonding device shown in FIG. **1**.

DESCRIPTION OF THE EMBODIMENTS

(4) The disclosure provides a bonding device, which can prevent a plate assembly from being bent and deformed to suppress an increase in a conductive contact area, thereby improving the bonding

effect.
(5) The disclosure provides a bonding device, which is used to bond a plate assembly including a first plate and a second plate configured to abut against the first plate and includes: an anvil, configured to support a first surface of the plate assembly provided by the first plate; a probe, disposed at a position corresponding to the anvil to be opposite to a second surface of the plate assembly provided by the second plate; a shoulder component, disposed around the probe and having a via for the probe to pass through and an abutting surface for pressing the second surface; a driving mechanism, configured to rotate the probe about a central axis intersecting the second surface and to move the probe forward or backward relative to the second plate along the central axis; a power supply, electrically connected to the anvil and the shoulder component, so that current flows through the plate assembly between the anvil and the shoulder component; and a control part, controlling operations of the driving mechanism and the power supply. The anvil has a hemispherical surface facing the first surface and being at least partially conductive and a receiving surface disposed at a position corresponding to the abutting surface of the shoulder component and being at least partially insulated, and the anvil contacts the first surface with the hemispherical surface and the receiving surface.
(6) In an embodiment of the disclosure, the anvil includes a handle part, a top part, and a receiving part. The handle part has a tapered portion at a front end, the top part has the hemispherical surface at a front end portion and has an engaging portion engaged with the tapered portion on an inner surface of a base end portion, and the receiving part surrounds an outer surface of the top part and overlaps with the tapered portion in a width direction and has the receiving surface at a front end portion.
(7) In an embodiment of the disclosure, the handle part has an edge portion located below the tapered portion, the top part is installed on the handle part, and a lower end portion of the top part abuts the edge portion of the handle part.
(8) In an embodiment of the disclosure, the receiving part includes a receiving portion and a holding portion. The receiving portion has the receiving surface contacting the first plate, the holding portion surrounds the receiving portion and is bonded to the handle part, and the receiving portion and the holding portion are separate components separately disposed.
(9) Based on the above, in the bonding device of the disclosure, the anvil has the hemispherical surface facing the first surface of the first plate and being at least partially conductive and the receiving surface disposed at the position corresponding to the abutting surface of the shoulder component and being at least partially insulated, and the anvil contacts the first surface with the hemispherical surface and the receiving surface. In this way, the anvil in the bonding device reduces the conductive contact area with the first plate via the hemispherical surface, while receiving a load from the shoulder component during a bonding process via the receiving surface, thereby preventing the conductive contact area between the plate assembly and the anvil from increasing (that is, preventing a current flow path from expanding) due to bending deformation of the plate assembly to sufficiently generate a nugget for bonding via energization. Accordingly, the bonding device of the disclosure can prevent bending deformation of the plate assembly to suppress the increase of the conductive contact area, thereby improving the bonding effect.
(10) Reference will now be made in detail to exemplary embodiments of the disclosure, examples of the exemplary embodiments are illustrated in the drawings. FIG. **1** is a cross-sectional schematic diagram of a bonding device for bonding a plate assembly including a first plate and a second plate according to an embodiment of the disclosure, FIG. **2** is a partially enlarged schematic diagram of the bonding device shown in FIG. **1** during a bonding process, and FIG. **3** is a cross-sectional schematic diagram of an anvil used in the bonding device shown in FIG. **1**. A bonding device **100** and an operation manner thereof of the embodiment will be described below with reference to FIG. **1** to FIG. **3**. In the following description, the bonding device **100** energizes a plate assembly P in a vertical direction for bonding, but the direction is merely exemplary, and the disclosure may be

applied to plate assemblies oriented in any expected direction and from any expected direction, such as laterally, upwardly, and obliquely, without departing from the scope of the disclosure. The disclosure does not limit the processing manner of bonding the plate assembly P by the bonding device **100**, which may be adjusted according to requirements.

(11) Please refer to FIG. **1**. In the embodiment, the bonding device **100** is used to bond the plate assembly P including a first plate P**1** and a second plate P**2** configured to abut against the first plate P**1**. The first plate P**1** and the second plate P**2** in the plate assembly P extend along a horizontal plane, and the plate assembly P may include more plates (for example, a middle plate P**3** sandwiched between the first plate P**1** and the second plate P**2**) according to requirements. Furthermore, the plates in the plate assembly P are all conductive plates made of metal materials. The first plate P**1** and the middle plate P**3** may be made of the same material or similar materials, and the second plate P**2** may be made of a different or dissimilar material. Furthermore, the materials of the first plate P**1** and the middle plate P**3** have higher strength and higher resistance than the material of the second plate P**2**. For example, the first plate P**1** and the middle plate P**3** are iron plates, and the second plate P**2** is an aluminum plate. However, the disclosure does not limit the number and the materials of plates in the plate assembly P, which may be adjusted according to requirements.

(12) Furthermore, in the embodiment, as shown in FIG. **1**, the bonding device **100** includes an anvil **110**, a probe **120**, a shoulder component **130**, a driving mechanism **140**, a power supply **150**, and a control part **160**. The anvil **110** is configured to support a first surface S**1** of the plate assembly P provided by the first plate P**1**. The probe **120** is disposed at a position corresponding to (for example, above) the anvil **110** to be opposite to a second surface S**2** of the plate assembly P provided by the second plate P**2**. The shoulder component **130** is disposed around the probe **120** (for example, is an annular component surrounding the probe **120**) and has a via **132** for the probe **120** to pass through and an abutting surface **134** for pressing the second surface S**2**. The driving mechanism **140** is configured to rotate the probe **120** about a central axis C intersecting the second surface S**2** and to move the probe **120** forward or backward relative to the second plate P**2** along the central axis C. The power supply **150** is electrically connected to the anvil **110** and the shoulder component **130**, so that current flows through the plate assembly P between the anvil **110** and the shoulder component **130**. The control part **160** controls the operations of the driving mechanism **140** and the power supply **150**. Furthermore, the anvil **110** has a hemispherical surface **110***a* facing the first surface S**1** and being at least partially conductive and a receiving surface **110***b* disposed at a position corresponding to the abutting surface **134** of the shoulder component **130** and being at least partially insulated, and the anvil **110** contacts the first surface S**1** with the hemispherical surface **110***a* and the receiving surface **110***b*.

(13) Specifically, in the embodiment, as shown in FIG. **1**, at least part (for example, a part corresponding to the hemispherical surface **110***a*) of the anvil **110** is made of a conductive material, so that the anvil **110** may abut the first surface S**1** of the plate assembly P provided by the first plate P**1** with the at least partially conductive hemispherical surface **110***a*. Furthermore, at least part (for example, a part corresponding to the receiving surface **110***b*) of the anvil **110** is made of an insulating material, so that the anvil **110** may abut the first surface S**1** of the plate assembly P provided by the first plate P**1** with the at least partially insulated receiving surface **110***b*. Accordingly, the probe **120** is disposed above the anvil **110** to correspond to the anvil **110** and extends along the central axis C. The probe **120** is made of a hard material and preferably has a cylindrical shape (with the center of circle corresponding to the central axis C). Furthermore, the probe **120** is connected to the driving mechanism **140**, so that the probe **120** may rotate about the central axis C and move forward or backward relative to the second plate P**2** along the central axis C (that is, move up and down in the vertical direction) via the drive by the driving mechanism **140**. However, the disclosure does not limit the specific structures of the anvil **110** and the probe **120**, which may be adjusted according to requirements.

(14) In addition, in the embodiment, as shown in FIG. **1**, the shoulder component **130** has a rotationally symmetrical shape about the central axis C and is disposed above the anvil **110** and surrounds the probe **120**. For the shoulder component **130**, a middle portion is provided with the via **132** and a lower end portion forms the abutting surface **134**. The shoulder component **130** may be made of a hard material, but not limited thereto. The via **132** is disposed along the central axis C and preferably has a circular hole shape. An inner diameter of the via **132** is greater than an outer diameter of the probe **120**. During a process of the probe **120** being driven via the driving mechanism **140** to move forward relative to the second plate P**2**, the probe **120** passes through the via **132** (as shown in FIG. **1**), and the via **132** and the probe **120** are concentrically separated. Furthermore, the probe **120** may also rotate about the central axis C. Here, although the probe **120** and the shoulder component **130** are illustrated as separate components, the shoulder component **130** may also be provided with a supporting structure for supporting the probe **120** (that is, the shoulder component **130** and the probe **120** are connected). Furthermore, the inner diameter of the via **132** may be only slightly greater than the outer diameter of the probe **120**. When the probe **120** rotates about the central axis C, an inner peripheral surface of the via **132** may be in sliding contact with an outer peripheral surface of the probe **120**. The disclosure does not limit the specific structure of the shoulder component **130** and the connection relationship between the shoulder component **130** and the probe **120**, which may be adjusted according to requirements.
(15) In addition, in the embodiment, as shown in FIG. **1**, the driving mechanism **140** includes a rotation driving mechanism **142** for rotating the probe **120** about the central axis C and a forward/backward driving mechanism **144** for moving the probe **120** forward or backward along the central axis C. Furthermore, the driving mechanism **140** performs control by the control part **160**, so that the probe **120** rotates about the central axis C and moves forward or backward along the central axis C. In addition, in other embodiments not shown, the anvil **110** and the shoulder component **130** may also be configured to move via the drive by the driving mechanism **140**, so that the movements of the anvil **110** and the shoulder component **130** may also be controlled by the control part **160**. Alternatively, the anvil **110** for supporting the plate assembly P and the shoulder component **130** for abutting the plate assembly P may also be moved or driven via other mechanisms. The disclosure does not limit the specific structure and the connection manner of the driving mechanism **140**, which may be adjusted according to requirements.
(16) Furthermore, in the embodiment, the power supply **150** is electrically connected to the anvil **110** and the shoulder component **130** (such as being each connected through a sliding contact mechanism or a flexible cable not shown). At least part (for example, the hemispherical surface **110***a*) of the anvil **110** is conductive, and at least part (for example, the abutting surface **134**) of the shoulder component **130** is conductive. Furthermore, the control part **160** controls the operation of the power supply **150**, so that current flows between the anvil **110** and the shoulder component **130**. In this case, since the hemispherical surface **110***a* of the anvil **110** contacts the first surface S**1** provided by the first plate P**1** and the abutting surface **134** of the shoulder component **130** presses the second surface S**2** provided by the second plate P**2**, and the hemispherical surface **110***a*, the abutting surface **134**, and the plate assembly P are conductive, current of the power supply **150** may be transmitted between the anvil **110** and the shoulder component **130** and flow through the plate assembly P to energize the plate assembly P, so as to bond the plate assembly P in a subsequent step. However, the disclosure does not limit the specific structures and the connection manner of the power supply **150** and the control part **160**, which may be adjusted according to requirements.
(17) It can be seen that in the embodiment, as shown in FIG. **1** and FIG. **2**, during a process of bonding the plate assembly P using the bonding device **100**, the plate assembly P is placed between the anvil **110** and the shoulder component **130**, so that the anvil **110** supports the first surface S**1** of the first plate P**1** and the shoulder component **130** presses the second surface S**2** of the second plate P**2** with the abutting surface **134**. As an example, the plate assembly P may be placed on the anvil **110** with the first surface S**1** facing downward, and the shoulder component **130** may press the

second surface S**2** with the abutting surface **134**. Thus, the anvil **110** squeezes the first surface S**1** of the plate assembly P from below, and the shoulder component **130** squeezes the second surface S**2** of the plate assembly P from above. The hemispherical surface **110***a* and the receiving surface **110***b* of the anvil **110** are substantially located at the same horizontal height to be able to simultaneously contact the first surface S**1** of the plate assembly P provided by the first plate P**1**. Furthermore, the hemispherical surface **110***a* of the anvil **110** is at least partially conductive to be used to energize the plate assembly P. Correspondingly, the receiving surface **110***b* of the anvil **110** is at least partially insulated to be used to bear a load applied from the shoulder component **130** to the plate assembly P (to be not electrically conducted with the plate assembly P to not affect bonding).

(18) Next, when the plate assembly P is pressed between the anvil **110** and the shoulder component **130**, the probe **120** rotates about the central axis C and moves forward relative to the second surface S**2** (that is, moves downward). An end portion of the probe **120** is inserted into the second plate P**2** and deforms the second plate P**2**. At this time, due to frictional heat generated by the rotating probe **120**, a part of the second plate P**2** surrounding the probe **120** becomes a plastic fluid and generates an annular plastic fluid region around the probe **120**. Once the end portion of the probe **120** reaches the middle plate P**3**, the control part **160** enables current provided by the power supply **150** to flow between the anvil **110** and the shoulder component **130**. When current flows between the anvil **110** and the shoulder component **130**, the probe **120** keeps rotating and moves further downward (forward), and the end portion of the probe **120** is inserted into the middle plate P**3**. As a result, resistance heat generated in the first plate P**1** and the middle plate P**3** and frictional heat generated by the rotation of the probe **120** generates a melting region (a region R**1** shown in FIG. **2**) between the first plate P**1** and the middle plate P**3**. At the same time, a part of the materials of the second plate P**2** and the middle plate P**3** near the probe **120** plastically flows. As a result, a part of the middle plate P**3** penetrates into the second plate P**2** (a region R**2** as shown in FIG. **2**) via the plastic flow.

(19) After the energizing operation is completed, the control part **160** controls the power supply **150** to stop supplying current between the anvil **110** and the shoulder component **130**, and controls the driving mechanism **140**, so that the probe **120** moves backward (that is, moves upward) along the central axis C while rotating. When current stops and the probe **120** moves away, the melting region (for example, the region R**1** of FIG. **2**) between the first plate P**1** and the middle plate P**3** solidifies to form a nugget, so that the first plate P**1** and the middle plate P**3** are firmly bonded to each other via the nugget formed by the melting region. Furthermore, a part of the middle plate P**3** penetrates into the second plate P**2** (for example, the region R**2** of FIG. **2**) via the plastic flow, so that the second plate P**2** and the middle plate P**3** are also firmly bonded to each other via the penetrated part. Subsequently, the anvil **110** and the shoulder component **130** move away from the plate assembly P, so that the bonded plate assembly P may be removed. However, the above description is only one operating process of the bonding device **100** of the embodiment. The operating process of the bonding device **100** may be appropriately adjusted according to the specific structure of the bonding device **100** and the number and the materials of plates of the plate assembly P to be bonded, and the disclosure is not limited thereto.

(20) It can be seen that in the embodiment, the anvil **110** has the hemispherical surface **110***a* facing the first surface S**1** of the first plate P**1** and being at least partially conductive and the receiving surface **110***b* disposed at the position corresponding to the abutting surface **134** of the shoulder component **130** and being at least partially insulated, and the anvil **110** contacts the first surface S**1** with the hemispherical surface **110***a* and the receiving surface **110***b*. That is, the anvil **110** contacts the first surface S**1** with the at least partially conductive hemispherical surface **110***a* to energize the plate assembly P and contacts the first surface S**1** with the at least partially insulated receiving surface **110***b* to bear the load applied to the plate assembly P by the abutting surface **134** of the shoulder component **130**. In this way, the anvil **110** in the bonding device **100** reduces a conductive

contact area with the first plate P**1** via the hemispherical surface **110***a*, while receiving the load from the shoulder component **130** during the bonding process via the receiving surface **110***b*, thereby preventing the conductive contact area between the plate assembly P and the anvil **110** from increasing (that is, preventing the current flow path from expanding) due to bending deformation of the plate assembly P to sufficiently generate the nugget for bonding via energization. Accordingly, the bonding device **100** can prevent the plate assembly P from being bent and deformed to suppress an increase in the conductive contact area, thereby improving the bonding effect.
(21) More specifically, in the embodiment, as shown in FIG. **3**, the anvil **110** includes a handle part **112**, a top part **114**, and a receiving part **116**. In other words, the anvil **110** is composed of at least the three parts, the handle part **112**, the top part **114**, and the receiving part **116**. The handle part **112** has a flow path **112***a* inside and has a tapered portion **112***b* at a front end (that is, an upper end). The top part **114** has the hemispherical surface **110***a* at a front end portion (that is, an upper end portion) and has an engaging portion **114***a* engaged with the tapered portion **112***b* on an inner surface of a base end portion (that is, a lower end portion). The receiving part **116** surrounds an outer surface of the top part **114** and overlaps with the tapered portion **112***b* in a width direction (that is, a left-right direction) and has the receiving surface **110***b* at a front end portion (that is, an upper end portion).
(22) It can be seen that in the embodiment, the handle part **112** is used as a base of the anvil **110**, so that the flow path **112***a* as a cooling structure for cooling water to flow is disposed inside, and the tapered portion **112***b* at the front end thereof is gradually inclined outward from top to bottom. Correspondingly, the top part **114** is installed on the handle part **112** to provide the at least partially conductive hemispherical surface **110***a* at the front end portion thereof. Therefore, the top part **114** may be made of a conductive material (that is, to be used as an electrode) or may be only partially conductive (for example, the hemispherical surface **110***a*), and the disclosure is not limited thereto. Furthermore, the engaging portion **114***a* disposed at the base end portion of the top part **114** is, for example, an edge part of an opening gradually expanding outward from top to bottom. In this way, the engaging portion **114***a* of the top part **114** may be slid along the tapered portion **112***b* and fixed onto the handle part **112** during an installing process, so that the top part **114** may be more easily installed on the handle part **112**. Furthermore, the receiving part **116** is formed in a cylindrical shape and is installed on an outer periphery of the handle part **112** and the top part **114** to provide the at least partially insulated receiving surface **110***b* at the front end portion thereof. Therefore, the receiving part **116** may be made of an insulating material or may be only partially insulated (by, for example, providing an insulating material on the receiving surface **110***b*), and the disclosure is not limited thereto. Furthermore, the receiving part **116** surrounds the outer surface of the top part **114** (with a small gap between the two), and a range of the outer surface of the top part **114** surrounded by the receiving part **116** includes the tapered portion **112***b*, so as to limit an installation position of the top part **114** (that is, to limit displacement of the engaging portion **114***a* engaged with the tapered portion **112***b*).
(23) Through the above arrangement, when the plate assembly P is pressed between the anvil **110** and the shoulder component **130** and the probe **120** penetrates into the plate assembly P, the load applied to the plate assembly P by the probe **120** and the shoulder component **130** is transmitted toward the hemispherical surface **110***a* (for example, provided by the top part **114**) and the receiving surface **110***b* (for example, provided by the receiving part **116**) of the anvil **110**. At this time, since the receiving part **116** surrounds the outer surface of the top part **114**, an outer periphery of the top part **114** is limited, so that the engaging portion **114***a* of the top part **114** does not further move downward along the tapered portion **112***b* of the handle part **112** and expand outward. Furthermore, the receiving surface **110***b* of the receiving part **116** may be used to receive most of the load from above (that is, the shoulder component **130** and the probe **120**). It can be seen from this that a conductive part (that is, the hemispherical surface **110***a*) and an insulating part (that is,

the receiving surface **110***b*) in the anvil **110** may be appropriately separately disposed and easily assembled. Furthermore, the anvil **110** suppresses displacement deformation (that is, suppresses a change in a processing position of the hemispherical surface **110***a*) of the top part **114** relative to the handle part **112** in a pressurizing direction (that is, downward) via the receiving part **116**, thereby sufficiently generating the nugget for bonding via energization. However, the disclosure is not limited thereto and may be adjusted according to requirements.

(24) Furthermore, in the embodiment, as shown in FIG. **3**, the handle part **112** has an edge portion **112***c* located below the tapered portion **112***b*, the top part **114** is installed on the handle part **112**, and a lower end portion **114***b* of the top part **114** abuts the edge portion **112***c* of the handle part **112**. In other words, in addition to using the cylindrical receiving part **116** to limit the installation position of the top part **114** from the outside, the installation position of the top part **114** may also be limited by providing the edge portion **112***c* below the tapered portion **112***b* of the handle part **112**. Since the edge portion **112***c* is located below the tapered portion **112***b* and extends outward compared to the tapered portion **112***b*, the edge portion **112***c* may serve as a moving end point of the engaging portion **114***a* of the top part **114** moving downward along the tapered portion **112***b*, and the lower end portion **114***b* of the top part **114** abuts the edge portion **112***c*. In this way, when the load applied to the plate assembly P by the probe **120** and the shoulder component **130** is transmitted toward the anvil **110**, since the lower end portion **114***b* of the top part **114** abuts the edge portion **112***c* of the handle part **112**, the engaging portion **114***a* of the top part **114** does not move further downward along the tapered portion **112***b* of the handle part **112** and expand outward. Thus, displacement deformation of the top part **114** relative to the handle part **112** in the pressurizing direction (that is, downward) can be suppressed (that is, the change in the processing position of the hemispherical surface **110***a* can be suppressed), thereby sufficiently generating the nugget for bonding via energization. However, the disclosure is not limited thereto and may be adjusted according to requirements.

(25) In addition, in the embodiment, as shown in FIG. **3**, the receiving part **116** includes a receiving portion **116***a* and a holding portion **116***b*. The receiving portion **116***a* has the receiving surface **110***b* contacting the first plate P**1**, the holding portion **116***b* surrounds the receiving portion **116***a* and is bonded to the handle part **112**, and the receiving portion **116***a* and the holding portion **116***b* are separate components separately disposed. In other words, the receiving part **116** is composed of at least the two parts, the receiving portion **116***a* and the holding portion **116***b*. The receiving portion **116***a* is formed in a cylindrical shape and surrounds the outer surface of the top part **114** (with a small gap between the two). Furthermore, the receiving portion **116***a* provides the receiving surface **110***b* on an outer periphery of the hemispherical surface **110***a*. Therefore, the receiving portion **116***a* may be made of an insulating material or may be only partially insulated (by, for example, providing an insulating material on the receiving surface **110***b*), and the disclosure is not limited thereto. Correspondingly, the holding portion **116***b* is formed in a cylindrical shape to surround an outer periphery of the receiving portion **116***a*. Furthermore, a lower side of the holding portion **116***b* surrounds and contacts the engaging portion **114***a* of the top part **114** to transmit current of the power supply **150** (as shown in FIG. **1**) to the hemispherical surface **110***a* of the top part **114**, and the lower side of the holding portion **116***b* is further bonded to (for example, screwed to) the handle part **112** for fixation. Therefore, the holding portion **116***b* may be made of a conductive material or may be only partially conductive, and the disclosure is not limited thereto.

(26) It can be seen that in the embodiment, an insulating part (that is, the receiving portion **116***a*) and a conductive part (that is, the holding portion **116***b*) in the receiving part **116** may be appropriately separately disposed and easily assembled. Furthermore, in a case where the receiving portion **116***a* is worn due to contact with the first plate P**1** of the plate assembly P, etc., the receiving portion **116***a* may be easily replaced. That is, it is not necessary to replace the entire receiving part **116**, but only the receiving portion **116***a* needs to be detached from the holding portion **116***b* and replaced with a new receiving portion **116***a*, so that the size of parts to be replaced

is greatly reduced to suppress the cost of consumables. However, the above description is only an example of the anvil **110**, and the disclosure does not limit the anvil **110** to be composed of the handle part **112**, the top part **114**, and the receiving part **116**, nor does the disclosure limit the receiving part **116** to be composed of the receiving portion **116***a* and the holding portion **116***b*, nor does the disclosure limit whether the flow path **112***a*, the tapered portion **112***b*, the engaging portion **114***a*, and the edge portion **112***c* are provided. As long as the anvil **110** in the bonding device **100** has the at least partially conductive hemispherical surface **110***a* to reduce the conductive contact area with the first plate P**1** and has the at least partially insulated receiving surface **110***b* to receive the load from the shoulder component **130** during the bonding process, the disclosure does not limit the specific structural composition of the anvil **110**, which may be adjusted according to requirements.

(27) In summary, in the bonding device of the disclosure, the anvil has the hemispherical surface facing the first surface of the first plate and being at least partially conductive and the receiving surface disposed at the position corresponding to the abutting surface of the shoulder component and being at least partially insulated, and the anvil contacts the first surface with the hemispherical surface and the receiving surface. In this way, the anvil in the bonding device reduces the conductive contact area with the first plate via the hemispherical surface, while receiving the load from the shoulder component during the bonding process via the receiving surface, thereby preventing the conductive contact area between the plate assembly and the anvil from increasing (that is, preventing the current flow path from expanding) due to bending deformation of the plate assembly to sufficiently generate the nugget for bonding via energization. Preferably, the anvil includes the handle part, the top part, and the receiving part. The handle part has the tapered portion at the front end, the top part has the engaging portion engaged with the tapered portion on the inner surface of the base end portion, and the receiving part surrounds the outer surface of the top part and overlaps with the tapered portion in the width direction. In this way, the anvil suppresses displacement deformation of the top part relative to the handle part in the pressurizing direction (that is, suppresses the change in the processing position of the hemispherical surface) via the receiving part, thereby sufficiently generating the nugget for bonding via energization. Accordingly, the bonding device of the disclosure can prevent bending deformation of the plate assembly to suppress the increase of the conductive contact area, thereby improving the bonding effect.

(28) Finally, it should be noted that the above embodiments are only used to illustrate, but not to limit, the technical solutions of the disclosure. Although the disclosure has been described in detail with reference to the above embodiments, persons skilled in the art should understand that the technical solutions described in the above embodiments may still be modified or some or all of the technical features thereof may be equivalently replaced. However, the modifications or replacements do not cause the essence of the corresponding technical solutions to deviate from the scope of the technical solutions of the embodiments of the disclosure.

Claims

1. A bonding device, used to bond a plate assembly comprising a first plate and a second plate configured to abut against the first plate, comprising: an anvil, configured to support a first surface of the plate assembly provided by the first plate; a probe, disposed at a position corresponding to the anvil to be opposite to a second surface of the plate assembly provided by the second plate; a shoulder component, disposed around the probe and having a via for the probe to pass through and an abutting surface for pressing the second surface; a driving mechanism, configured to rotate the probe about a central axis intersecting the second surface and to move the probe forward or backward relative to the second plate along the central axis; a power supply, electrically connected to the anvil and the shoulder component, so that current flows through the plate assembly between

the anvil and the shoulder component; and a control part, controlling operations of the driving mechanism and the power supply, wherein the anvil has a hemispherical surface facing the first surface and being at least partially conductive and a receiving surface disposed at a position corresponding to the abutting surface of the shoulder component and being at least partially insulated, and the anvil contacts the first surface with the hemispherical surface and the receiving surface.

2. The bonding device according to claim 1, wherein: the anvil comprises a handle part, a top part, and a receiving part, the handle part has a tapered portion at a front end, the top part has the hemispherical surface at a front end portion and has an engaging portion engaged with the tapered portion on an inner surface of a base end portion, and the receiving part surrounds an outer surface of the top part and overlaps with the tapered portion in a width direction and has the receiving surface at a front end portion.

3. The bonding device according to claim 2, wherein: the handle part has an edge portion located below the tapered portion, the top part is installed on the handle part, and a lower end portion of the top part abuts the edge portion of the handle part.

4. The bonding device according to claim 2, wherein: the receiving part comprises a receiving portion and a holding portion, the receiving portion has the receiving surface contacting the first plate, the holding portion surrounds the receiving portion and is bonded to the handle part, and the receiving portion and the holding portion are separate components separately disposed.
